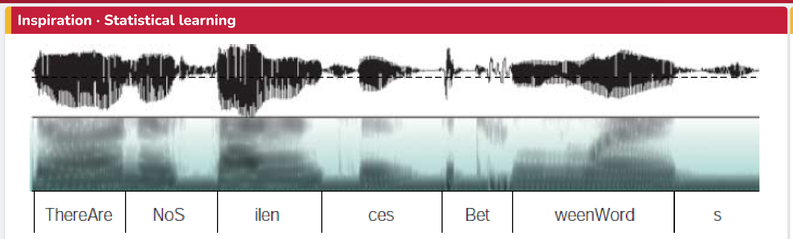
Emergent lexical features and network structural remodeling in an RNN trained on next-token prediction for a statistical learning task

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Code & figures on GitHub

Inspiration · Statistical learning



bupadokutabakutababupadokutababupado

bupado ? padoku

word non-word

Aims

- Is an RNN trained on next-token prediction able to learn words from an unsegmented stream?
- How is lexical information (word position, character, etc.) represented in a recurrent network that learns to produce words?
- How does the network's representation of information change with vocabulary complexity?
- How do the network's weights differ as a function of the complexity of the learned vocabulary (function → structure)?

Fig 1 · Mixed vocabularies & minimal DFAs

Vocabulary A · DFA=6

bat · bake · bank

batbakebankbatbankbakebatbakebank

Vocabulary B · DFA=8

hat · hate · late

hatlatehatehatlatehatehathatelatehat

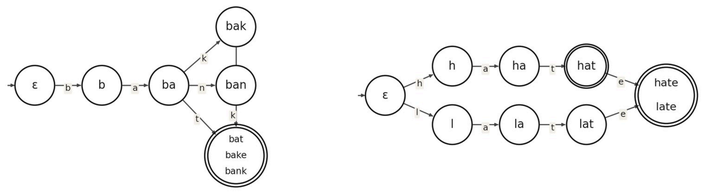
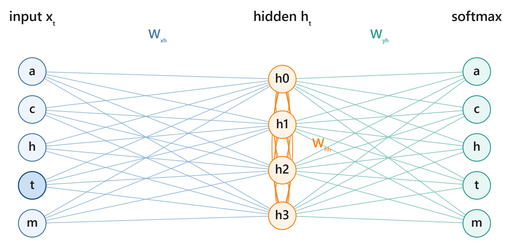


Fig 2 · Training the RNN



$h_t = \tanh(W_{xh} x_t + W_{hh} h_{t-1})$ · predict next char · cross-entropy loss

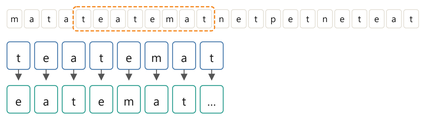
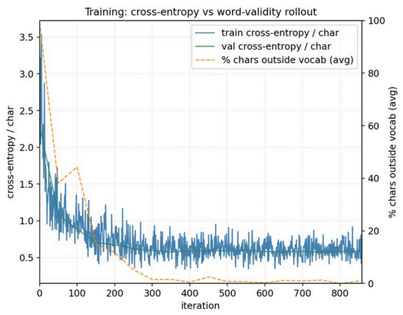


Fig 3 · Learning & generation



Generation before vs after learning

ate, cat, gate, hat, late, mat, plate, slate

Before training

high out-of-vocabulary rate

cepsgtthcmgpecalseptmgethcshptgmmchhcapcet

After training

mostly vocabulary words

atcatcatgateplatemathatmatplateplatelatesl

in-vocabulary out-of-vocabulary

Fig 4 · Next-character probabilities

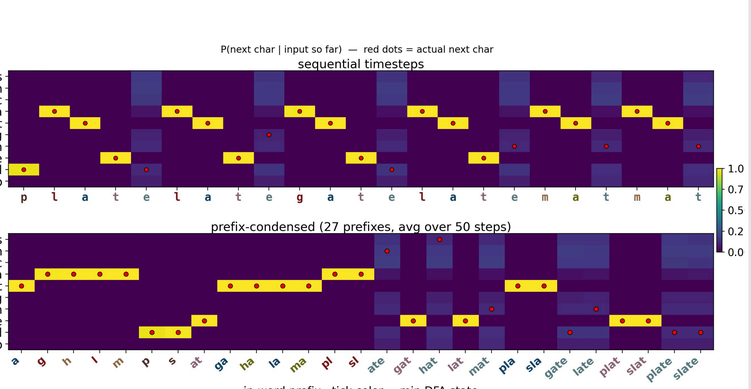


Fig 5 · Activation heatmap

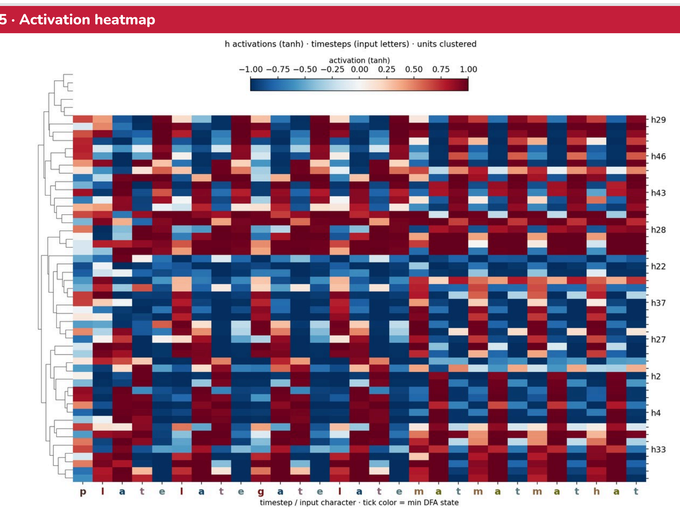


Fig 6 · Activations by prefix

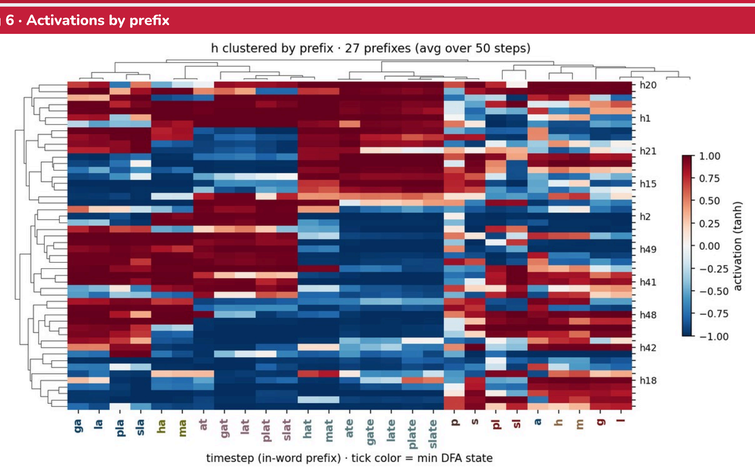


Fig 7 · Hidden-state correlation

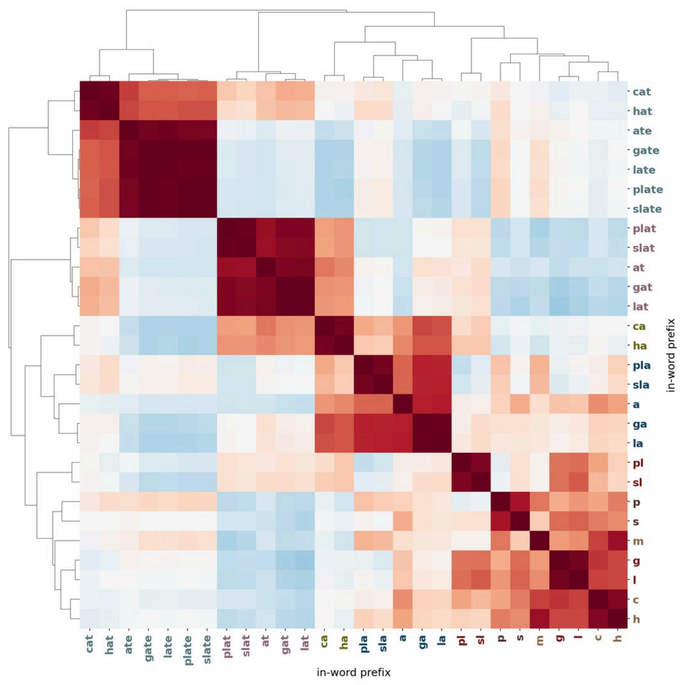


Fig 8 · DFA geometry & separation

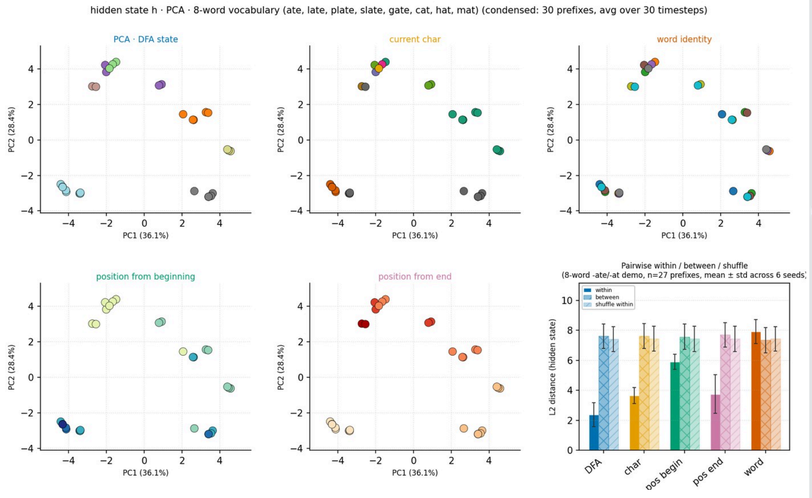


Fig 9 · Example selective units, prior to learning

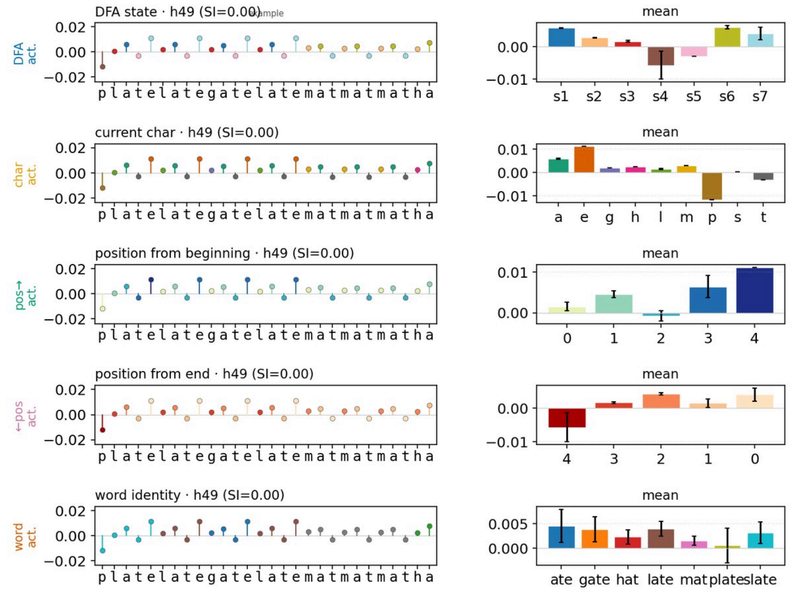


Fig 10 · Linear decoding, prior to learning

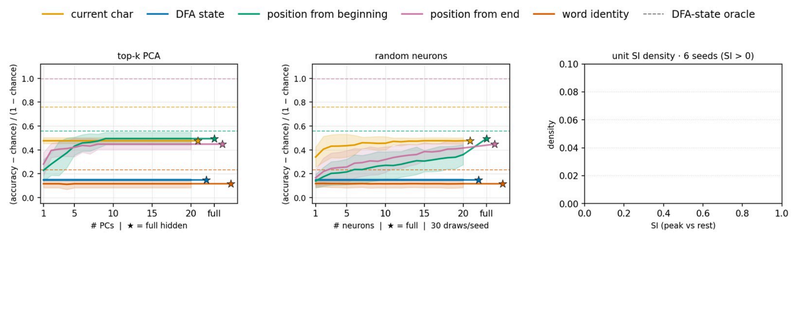


Fig 11 · Word trajectories

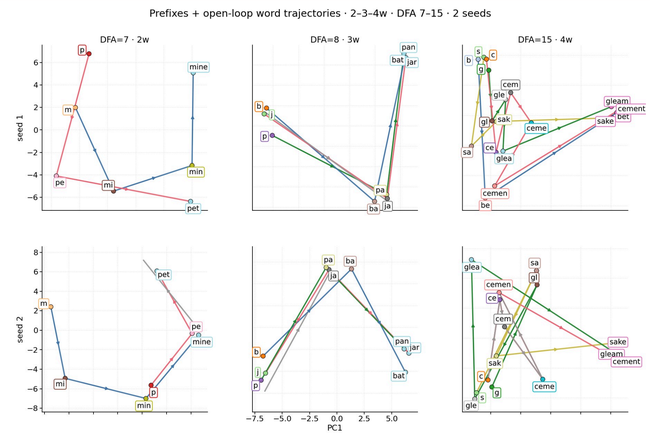


Fig 12 · Activations & scaling vs DFA size

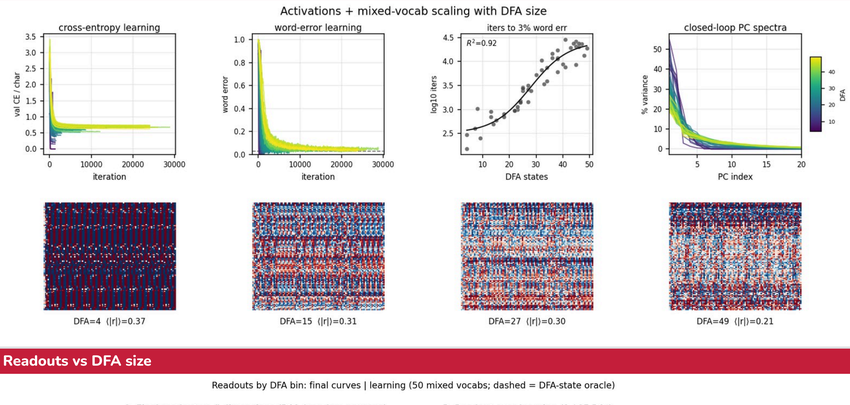


Fig 13 · Readouts vs DFA size

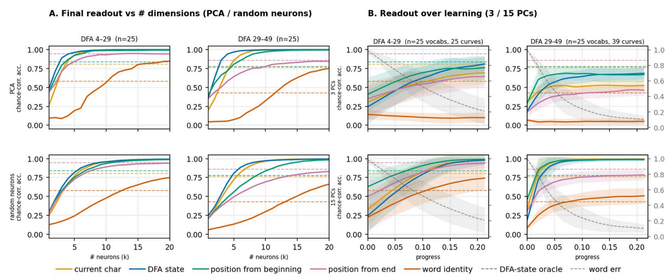


Fig 14 · Geometry & unit selectivity

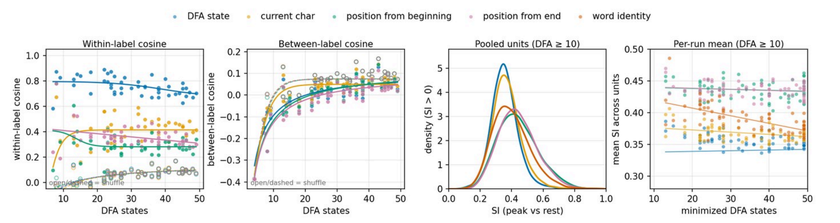


Fig 15 · Weight matrices vs DFA

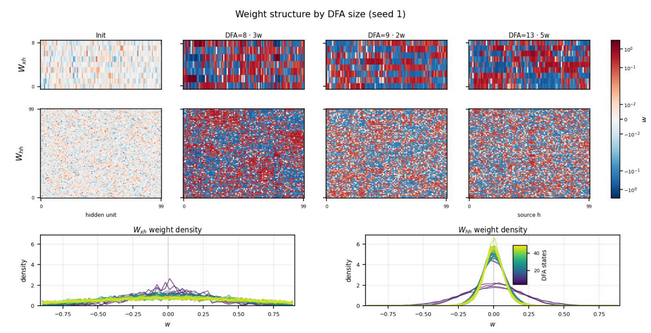
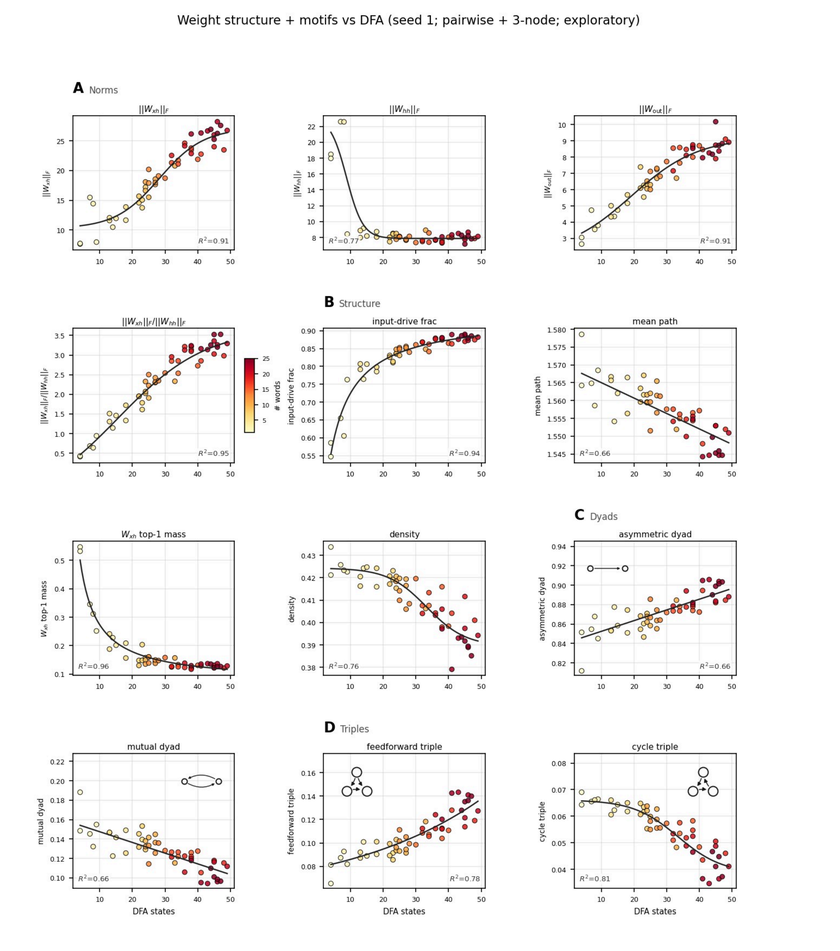


Fig 16 · Weight graph motifs vs DFA



Contact: Toviah Moldwin

Key references: Saffran, Aslin & Newport (1996); Elman (1990); Aslin, Saffran & Newport (1998); Goldwater et al. (2009); Perruchet & Vinter (1998).